Instrumentation:

Temperature Sensors- Part1

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- ✓ معرفی ترموکوپل (انواع و روش استفاده)
- ✓ اثر گرمامقاومتی در فلزات (پدیده ی RTDها)
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انواع حسگرهای دما

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۲- حسگرهای دمای مقاومتی

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- ✓ ترمیستور

۳- حسگرهای دمایی مدار مجتمع

۴- حسگرهای دیداری مادون قرمز (اندازه گیری به روش غیر مستقیم)

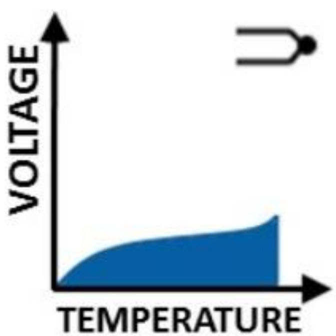
مقایسه حسگرهای دما

مقایسه محدوده دمایی:

Sensor	Useful Temperature Range
Thermocouples	**–270°C to 2320°C**
Typical resistance temperature detectors (RTDs)[1]	–196°C to 661°C
Thermistors	–55°C to 100°C
Integrated circuit sensors	–55°C to 150°C

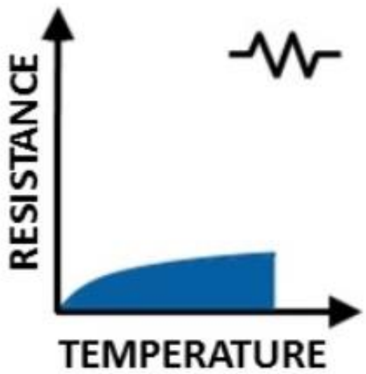
Ref: T. W. Kerlin ,and M. Johnson. *Practical thermocouple thermometry*. Instrument Society of America, 2012.

مزایا و معایب ترموکوپل:

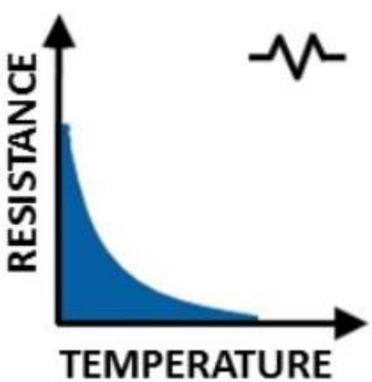
	Advantages	**Disadvantages**
THERMOCOUPLES  	✓ Simple ✓ Rugged ✓ Inexpensive ✓ No external power ✓ Wide temperature range ✓ Variety of styles	× Nonlinear response × Small sensitivity × Small output voltage × Requires CJC **(Cool Junction Compensation)** × Least stable

مقایسه حسگرهای دما

مزایا و معایب RTD:

RTD 	✓ Most stable ✓ Good Linearity ✓ Most accurate	× Low sensitivity × Externally powered × Costly × Small output resistance × Self-heating error

مزایا و معایب Thermistor:

THERMISTOR 	✓ Fast ✓ High output ✓ Minimal lead resistance error	× Limited temperature range × Externally powered × Nonlinear × More fragile × Self-heating error

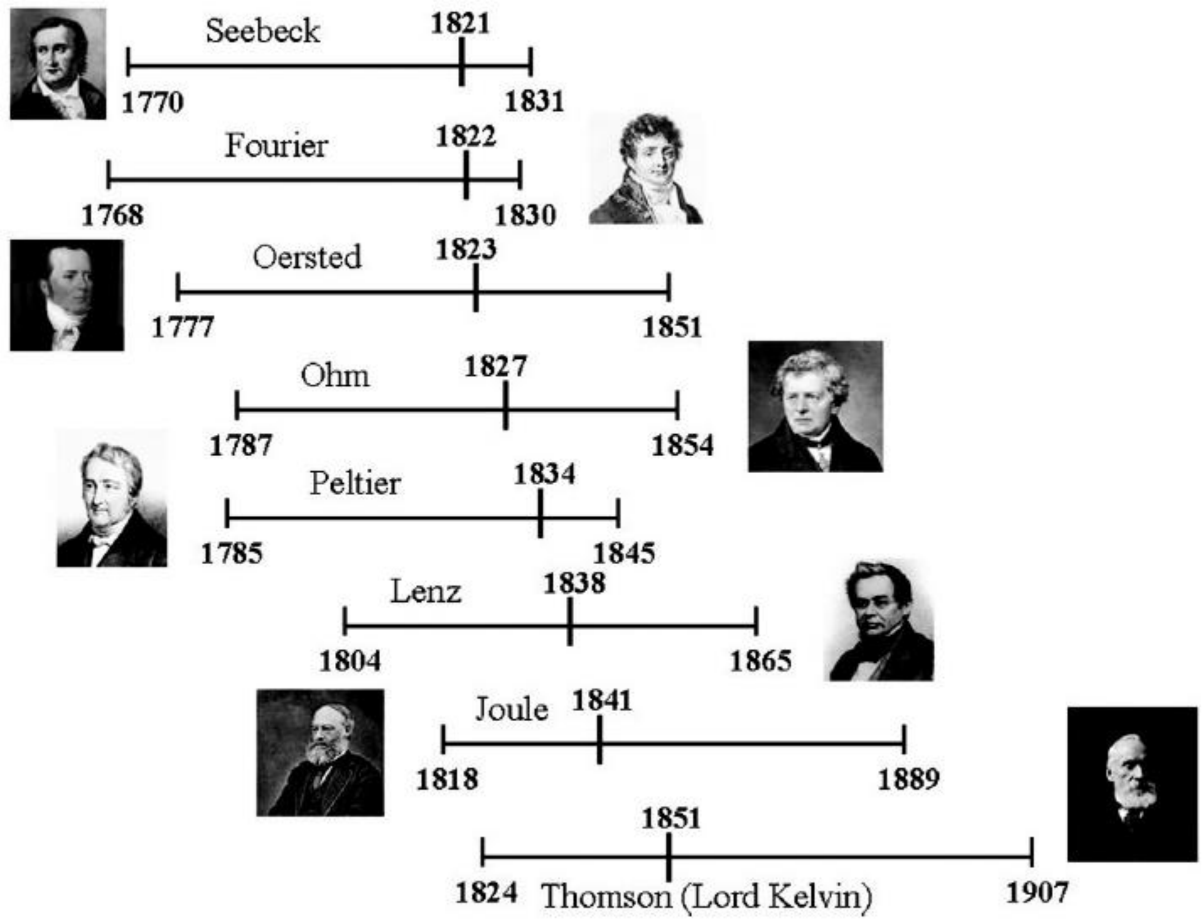
National Instruments Company -
http://sine.ni.com/np/app/main/p/ap/daq/lang/en/pg/1/sn/n17:daq,n21:17557/fmid/6681/

مقایسه حسگرهای دما

Characteristic	Thermocouple	RTD	Thermistor
Temperature Range	Excellent -210 °C to 1760 °C	Great -240 °C to 650 °C	Good -40 °C to 250 °C
Linearity	Fair	Good	Poor
Sensitivity	Low	Medium	Very High
Response Time	Medium to Fast	Medium	Medium to Fast
Stability	Fair	Good	Poor
Accuracy	Medium	High	Medium
Susceptible to Self-Heating?	No	Yes, Minimal	Yes, Highly
Durability	Excellent	Good	Poor
Cost	Lowest	High	Low
Signal Conditioning Requirements	Cold-Junction Compensation Amplification Open Thermocouple Detection Scaling	Excitation Lead Resistance Correction Scaling	Excitation Scaling

Table 2: *Comparison of Temperature Sensor Types*

Thermoelectric effects: The phenomena linking thermal energy transport and electrical currents in solid materials are collectively known as *thermoelectric effects*.



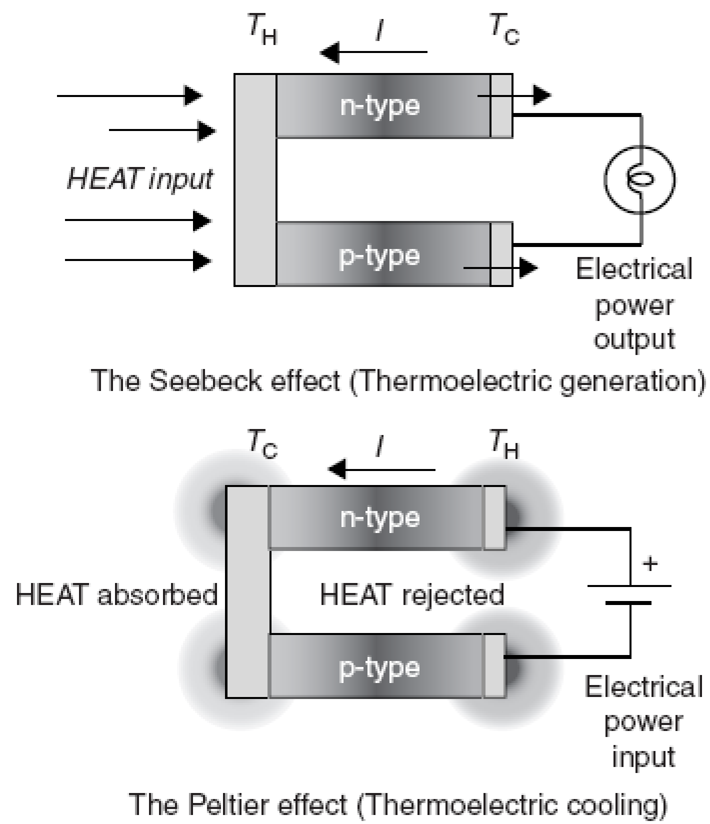
Chronogram showing the portraits and life span of the main characters in the origins of thermoelectric research. The ticks indicate the date when the corresponding thermoelectric phenomenon was first reported.

Seebeck Effect

Peltier Effect

Thomson Effect

The Kelvin Relationships



Thermoelectric generator (upper); thermoelectric refrigerator (lower).

The thermometric effects which underlie thermoelectric energy conversion can be conveniently discussed with reference to the schematic of a thermocouple shown in Figure 1.2. It can be considered as a circuit formed from two dissimilar conductors, a and b (referred to in thermoelectrics as thermocouple legs, arms, thermoelements, or simply elements and sometimes as pellets by device manufacturers) which are connected electrically in series but thermally in parallel. If the junctions at A and B are maintained at different temperatures T_1 and T_2 and $T_1 > T_2$ an open circuit electromotive force (emf), V is developed between C and D and given by $V = \alpha(T_1 - T_2)$ or $\alpha = V/\Delta T$, which defines the differential Seebeck coefficient α_{ab} between the elements a and b. For small temperature differences the relationship is linear. Although by convention α is the symbol for the Seebeck coefficient, S is also sometimes used and the Seebeck coefficient referred to as the thermal emf or thermopower. The sign of α is positive if the emf causes a current to flow in a clockwise direction around the circuit and is measured in V/K or more often in $\mu V/K$.

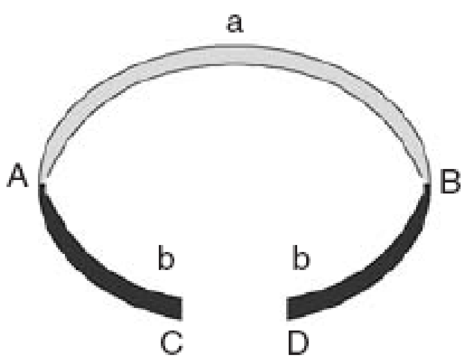
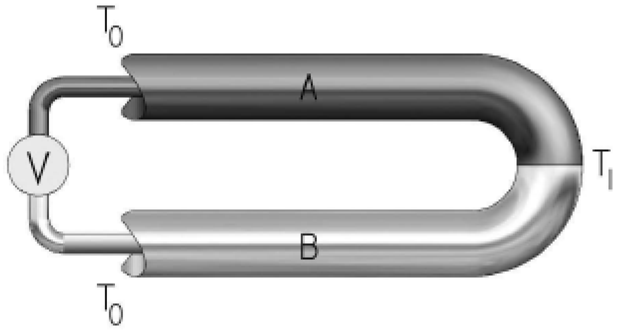
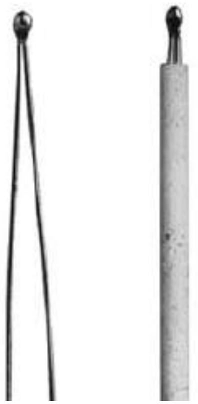


FIGURE 1.2 Schematic basic thermocouple.



Schematic View of a Thermocouple



(a) Conventional metal metal/alloy thermocouple

[Thermoelectrics handbook: macro to nano. CRC press, 2005]

$$V = S_A(T_1 - T_0) + S_B(T_0 - T_1)$$

contribution from conductor A — *contribution from conductor B*

$$V = S_A(T_1 - T_0) - S_B(T_1 - T_0)$$

$$V = (S_A - S_B)(T_1 - T_0)$$

$$S_{AB} = S_A - S_B$$

$$V = S_{AB}(T_1 - T_0)$$

Table 1.1 Seebeck coefficient values of different materials at $T = 273$ K

Metal	S (μVK^{-1})	Metal	S (μVK^{-1})
Ni	−18.0	Pd	−9.00
Pt	−4.45	Pb	−1.15
V	+0.13	W	+0.13
Rh	+0.48	Ag	+1.38
Cu	+1.70	Au	+1.79
Mo	+4.71	Cr	+18.0

[Maciá, Enrique, ed. Thermoelectric Materials: Advances and Applications. CRC Press, 2015]

If in Figure 1.2 the reverse situation is considered with an external emf source applied across C and D and a current I flows in a clockwise sense around the circuit then a rate of heating q occurs at one junction between a and b and a rate of cooling $-q$ occurs at the other. The ratio of I to q defines the Peltier coefficient given by $\pi = I/q$, is positive if A is heated and B is cooled, and is measured in watts per ampere or in volts.

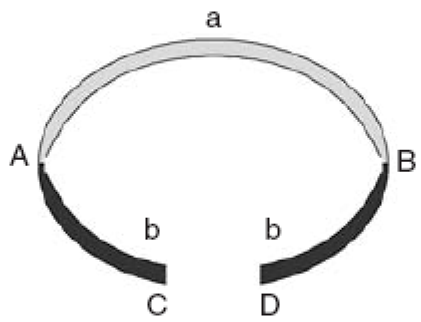
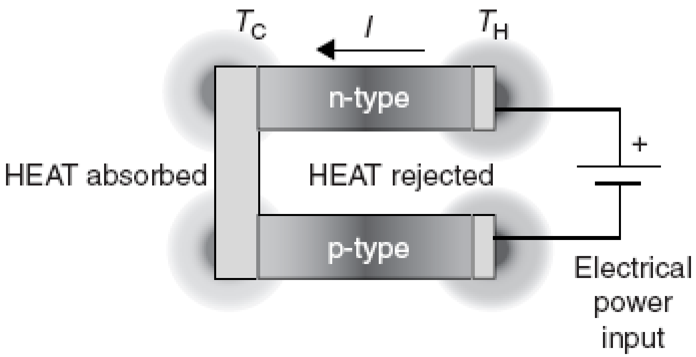
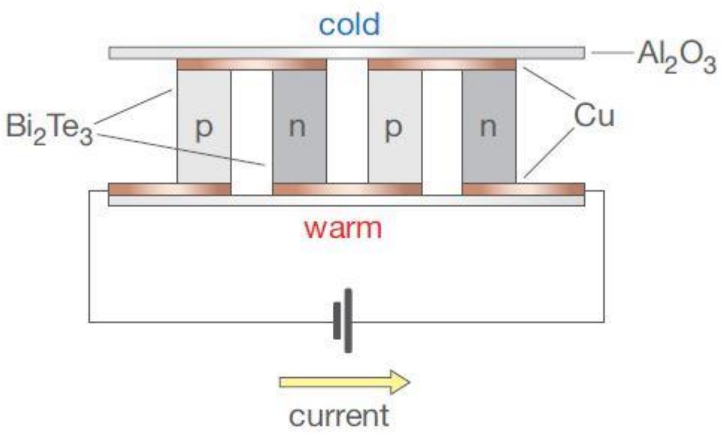


FIGURE 1.2 Schematic basic thermocouple.



The Peltier effect (Thermoelectric cooling)

نمونه های از سردکن پلتیر



https://www.azom.com/article.aspx?ArticleID=11389

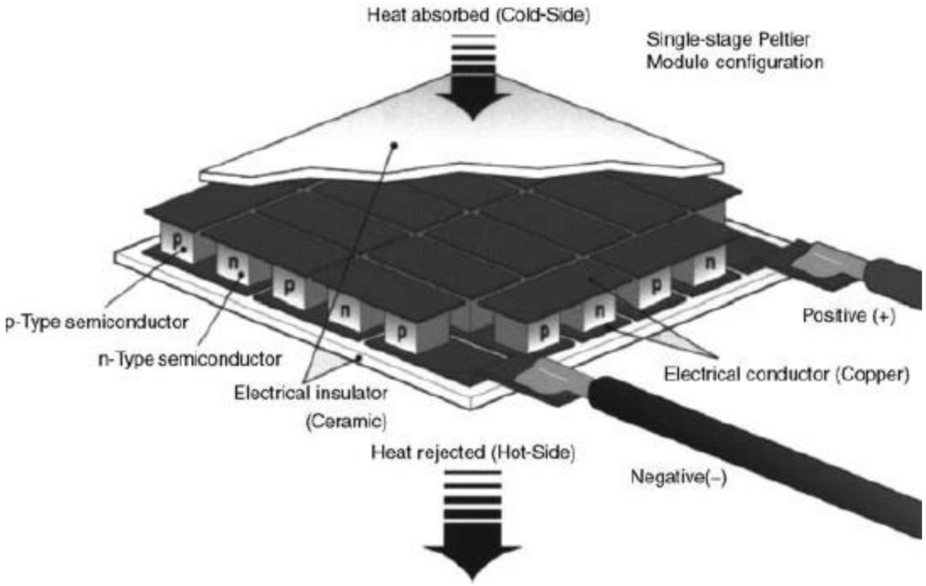
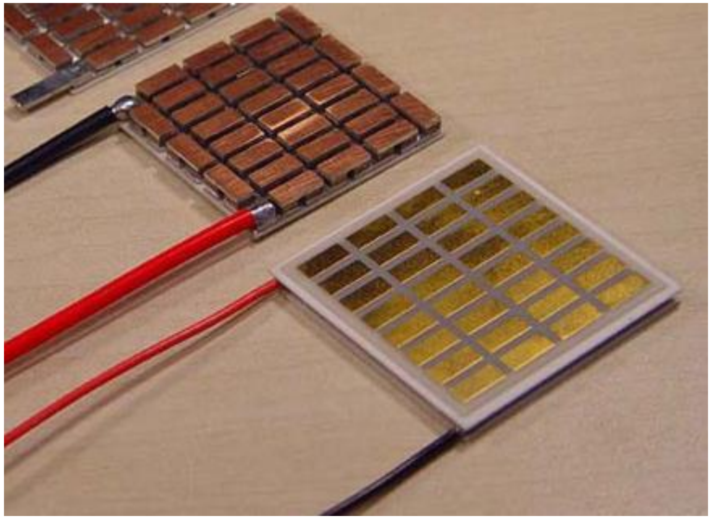


FIGURE 1.11 Cutaway of a typical Peltier cooler.

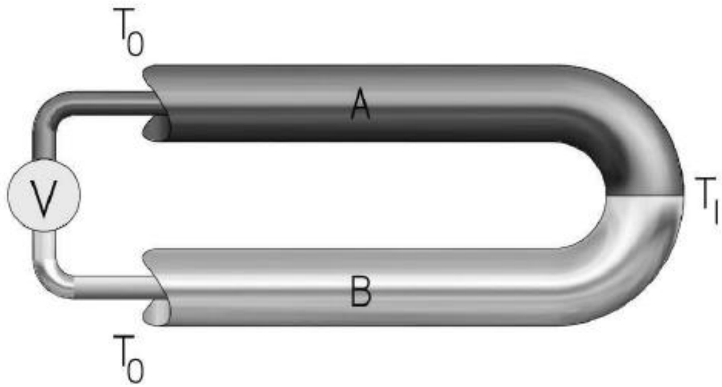


http://www.huimao.com/product/showproduct.php?id=23&lang=en

Rowe, David Michael, ed. Thermoelectrics handbook: macro to nano. CRC press, 2005.

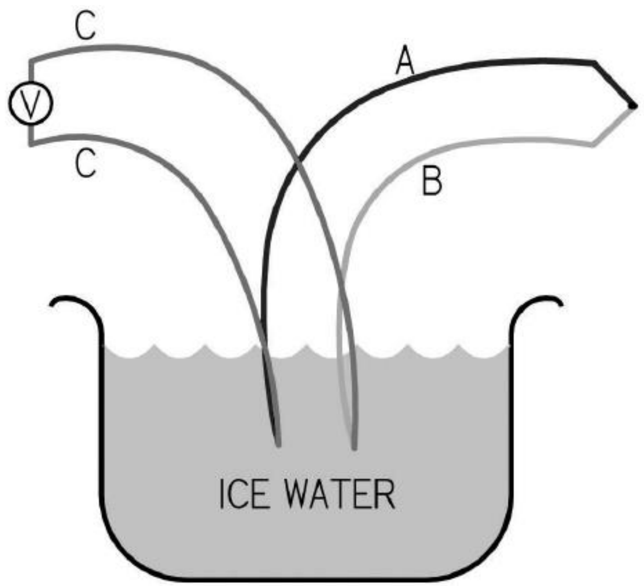
$$V = S_A(T_1 - T_0) + S_B(T_0 - T_1)$$

contribution from conductor A — *contribution from conductor B*

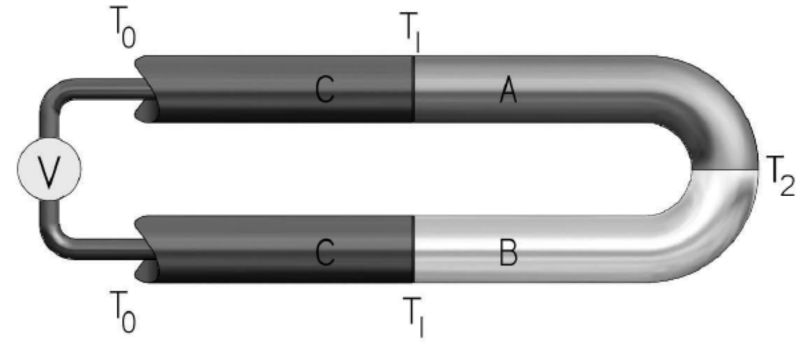


The Law of Thermocouple Thermometry: The emf produced by a segment of parallel dissimilar wires that experiences a temperature difference across the segment is proportional to the temperature difference. The total emf produced by the total circuit is the algebraic sum of the emfs produced by each segment between the open end and the junction of the wires.

ایجاد دمای مرجع برای ترموکوپل



Thermocouple with Ice Bath Reference

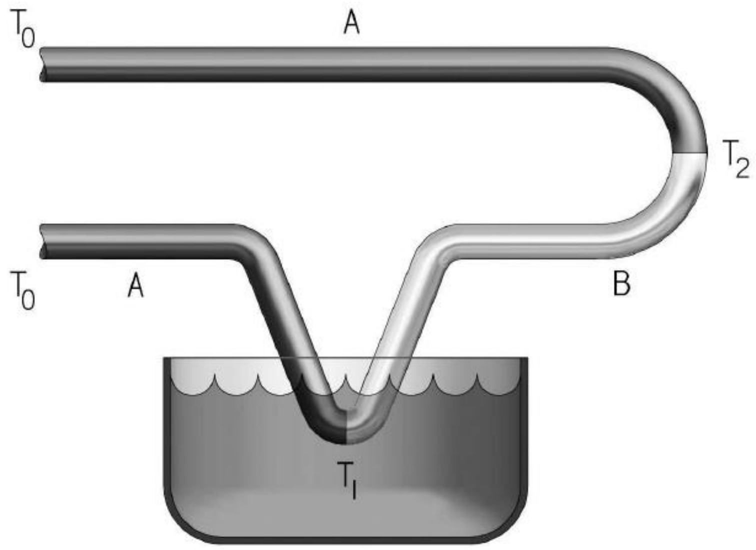


A Thermocouple with Identical Extension Wires

$$V = S_C(T_1 - T_0) + S_A(T_2 - T_1) + S_B(T_1 - T_2) + S_C(T_0 - T_1)$$

$$V = S_{AB}(T_2 - T_1)$$

ایجاد دمای مرجع برای ترموکوپل



An Alternate Ice Bath Reference

$$V = S_A(T_2 - T_0) + S_B(T_1 - T_2) + S_A(T_0 - T_1)$$

$$V = S_A(T_2 - T_1) + S_B(T_1 - T_2)$$

$$V = S_{AB}(T_2 - T_1)$$

$$°F = 1.8 \times °C + 32$$

$$°C = (°F - 32)/1.8$$

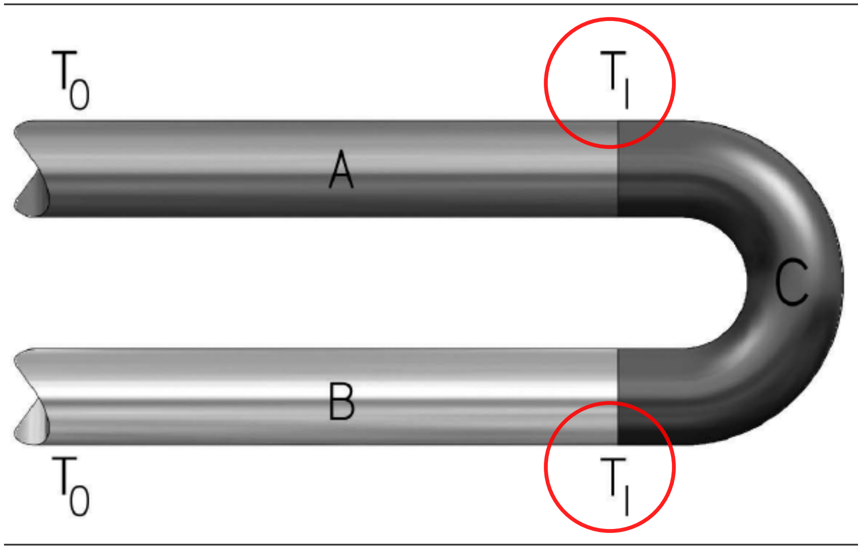
$$K = °C + 273.15$$

$$°R = °F + 459.67$$

Celsius and Fahrenheit scales are related to the Kelvin and Rankine absolute scales, respectively.

نحوه اتصال ترموکوپل

All of the depictions of thermocouples in previous sections have showed the two thermocouple wires joined at the junction, but there was no mention of *how* they were joined. Were they twisted together, welded, soldered, bolted, clamped—or what? *Thermoelectrically, it does not matter*!



A Thermocouple with a Third Material at the Junction

هم دمایی در هر دو سر مواد ترموالکتریک مهم است.

$$V = S_A(T_1 - T_0) + S_C(T_1 - T_1) + S_B(T_0 - T_1)$$

$$V = S_{AB}(T_1 - T_0)$$

نکات مهم در اندازه گیری بوسیله ترموکوپل

- Voltage is not produced at the junction of the thermocouple wires.
- Voltage is produced along the portions of the thermocouple wires that experience temperature differences.
- Voltage for an ideal thermocouple is related to the temperature difference between the junction end and the open end.
- Thermocouple loop analysis is simple and can explain all the important phenomena in thermocouples related to temperature measurement. Even casual users of thermocouples will benefit by understanding and using this simple analysis method.
- For temperature measurement, the quantity of interest is the open-circuit voltage (OCV), that is, the voltage that occurs when there is no current flowing.
- It does not matter how thermocouple wires are joined (twisted, welded, soldered, bolted, clamped, etc.) insofar as the thermocouple's temperature measuring capability is concerned.

T. W. Kerlin and M. Johnson. *Practical thermocouple thermometry*. Instrument Society of America, 1999.

Table 2-1. ASTM Thermocouple Types

Type	Principle Wire Constituents
J	Iron vs. nickel-copper alloy
T	Copper vs. nickel-copper alloy
K	Nickel-chromium alloy vs. nickel-manganese-silicon-aluminum alloy
E	Nickel-chromium alloy vs. nickel-copper alloy
N	Nickel-chromium-silicon alloy vs. nickel-silicon-magnesium alloy
C	Tungsten-rhenium alloy vs. tungsten-rhenium alloy
S	Platinum-rhodium alloy vs. platinum
R	Platinum-rhodium alloy vs. platinum
B	Platinum-rhodium alloy vs. platinum-rhodium alloy

American Society for Testing and Materials (ASTM)

انواع ترموکوپل

The reference temperature of 0°C.

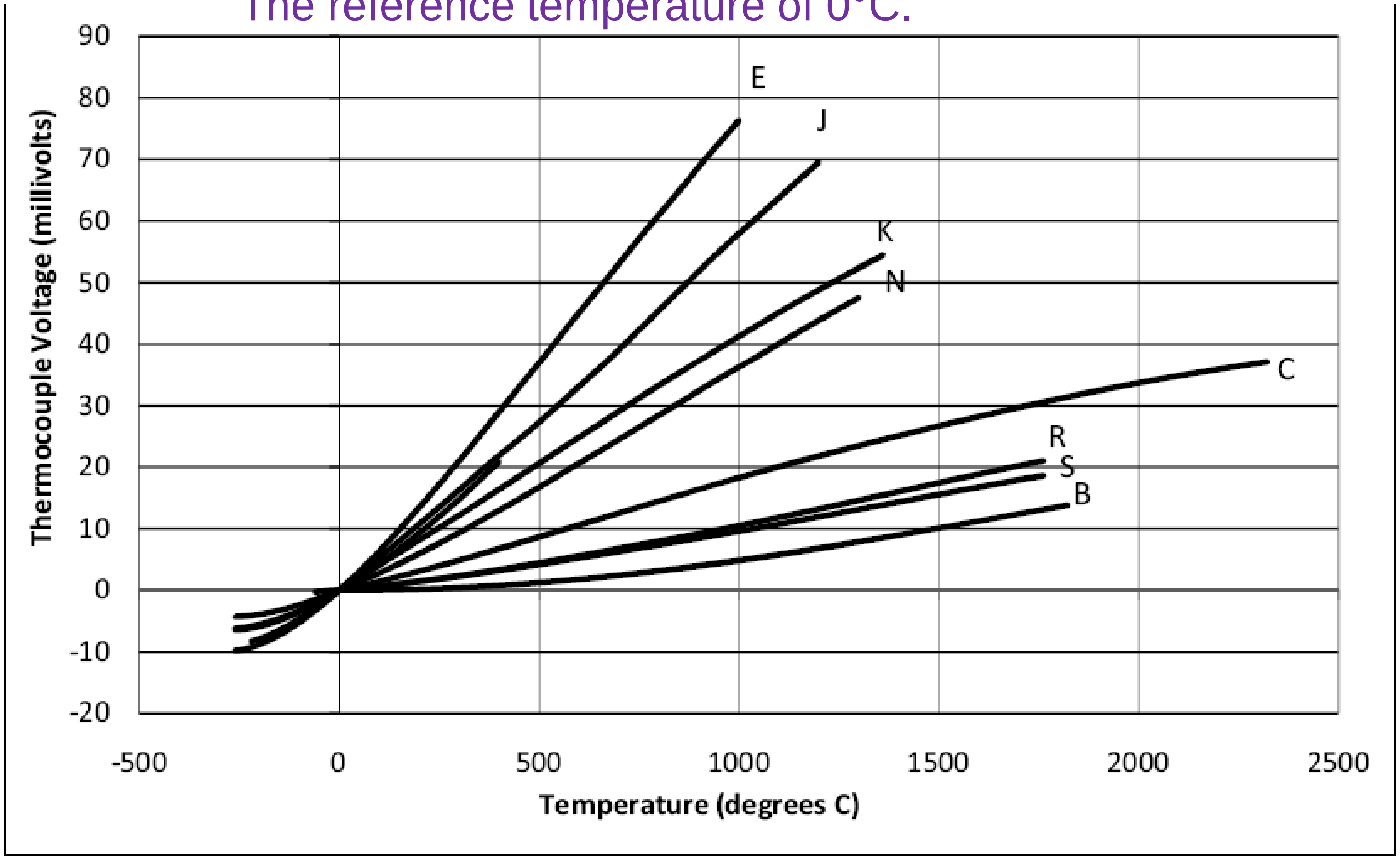


Figure 3-1. Thermoelectric EMFs for Standard Thermocouples

Converting EMF to Temperature

Now, let us consider the situation in which the reference-end temperature is not 0°C but is known. If the known temperature is T_1, then we can write

$$\boxed{V(0°\text{C} \rightarrow T_2) = V(0°\text{C} \rightarrow T_1) + V(T_1 \rightarrow T_2)}$$

$V(0°C \rightarrow T_2) =$ voltage produced by the thermocouple with the reference end at 0°C and the measuring junction at temperature T_2

$V(0°C \rightarrow T_1) =$ voltage produced by the thermocouple with the reference end at 0°C and the measuring junction at temperature T_1

$V(T_1 \rightarrow T_2) =$ voltage produced by the thermocouple with the reference end at temperature T_1 and the measuring junction at temperature T_2

EXAMPLE

A Type N thermocouple produces an emf of 10.610 mV when the open-end temperature is 20°C. What is the measuring-junction temperature?

SOLUTION

According to Appendix C, $V(0°C \rightarrow 20°C)$ is 0.525 mV.

Therefore,

$$V(0°C \rightarrow T_2) = 0.525 + 10.610 = 11.135\ mV$$

This is the emf that would have been measured if the reference temperature had been 0°C. Again, using Appendix C, we find that $T_2 = 350°C$.

TYPE N

emf in Millivolts — Reference Junctions at 0°C

°C	0	-10	-20	-30	-40	-50	-60	-70	-80	-90	-100
-200	-3.990	-4.083	-4.162	-4.226	-4.277	-4.313	-4.336	-4.345			
-100	-2.407	-2.612	-2.808	-2.994	-3.171	-3.336	-3.491	-3.634	-3.766	-3.884	-3.990
0	0.000	-0.260	-0.518	-0.772	-1.023	-1.269	-1.509	-1.744	-1.972	-2.193	-2.407
°C	**0**	**10**	**20**	**30**	**40**	**50**	**60**	**70**	**80**	**90**	**100**
0	0.000	0.261	0.525	0.793	1.065	1.340	1.619	1.902	2.189	2.480	2.774
100	2.774	3.072	3.374	3.680	3.969	4.302	4.618	4.937	5.259	5.585	5.913
200	5.913	6.245	6.579	6.916	7.255	7.597	7.941	8.288	8.637	8.968	9.341
300	9.341	9.696	10.054	10.413	10.774	11.136	11.501	11.867	12.234	12.603	12.974

EXAMPLE

A Type J thermocouple is connected to copper wires that connect to a readout instrument. What voltage is produced if the junction is at 400°C and the connection to copper is at 100°C?

SOLUTION

The copper section contributes no voltage because both conductors are identical. The Type J segment contributes the following voltage:

$$V = V(400^oC - 100^oC)$$

Using Equation 2-10 gives

$$V(400^oC - 100^oC) = V(400^oC - 0^oC) - V(100^oC - 0^oC)$$

That is, we can use the thermocouple tables (referenced to 0°C). Using the table in Appendix C for Type J thermocouple gives

$$V(400^oC - 100^oC) = 21.848 - 5.269 = 16.579\ mv$$

TYPE J

Reference Junctions at 0°C

emf in Millivolts

°C	0	10	20
0	0.000	0.507	1.019
100	5.269	5.814	6.360
200	10.779	11.334	11.889
300	16.327	16.881	17.434
400	21.848	22.400	22.952
500	27.393	27.953	28.516

Thermoresistive Effects

The electrical resistance of a metal conductor increases as temperature increases. This is because the electrical conductivity of a metal relies on the movement of electrons through its crystal lattice. Due to thermal excitation, the vibration of electrons increases, which slows the electrons' movement, thus causing the resistance to increase. The relationship between resistance R and temperature T can be expressed by a polynomial equation:

$$R_T = R_0[1 + A(T - T_0) + B(T - T_0)^2 + C(T - T_0)^3 + \cdots] \quad (2.6)$$

Its simplified version is

$$R_T = R_0[1 + A(T - T_0)] \quad (2.7)$$

where R_0 is resistance at the reference temperature T_0 (usually either 0°C, 20°C, or 25°C); $A, B, C, \ldots$ are material-dependent temperature coefficients (in $\Omega \cdot \Omega^{-1} \cdot {}^\circ C^{-1}$).

اثر گرمامقاوتی در فلزات (اساس کارکرد RTDها)

Metals have *positive temperature coefficients* (PTC), because their resistance increases as the temperature increases. All resistance temperature devices (RTDs), made of metals, are PTC sensors. The temperature coefficient for all pure metals is of the order of 0.003–0.007 $\Omega \cdot \Omega^{-1} \cdot {}^{\circ}C^{-1}$. Temperature coefficients *A* for common metals are listed in Table 2.2 [2].

Common Metals' Temperature Coefficients at 20°C

Metal	Temperature Coefficients *A* at 20°C ($\Omega \cdot \Omega^{-1} \cdot {}^{\circ}C^{-1}$)
Gold	0.003715
Silver	0.003819
Copper	0.004041
Aluminum	0.004308
Tungsten	0.004403
Iron	0.005671
Nickel	0.005866

Source: From Temperature coefficient of resistance, Creative Commons, Stanford, California, USA, 2008. With permission.

EXAMPLE 2.4

A copper wire has a resistance of 5 Ω at 20°C. Calculate its resistance if the temperature is increased to 65°C.

SOLUTION

From Table 2.2, $A = 0.004041\ \Omega \cdot \Omega^{-1} \cdot {}^\circ\mathrm{C}^{-1}$ at 20°C. Use Equation 2.7:

$$R_T = R_0[1 + A(T - T_0)] \Rightarrow R_{65} = (5\ \Omega)[1 + 0.004041\ \Omega \cdot \Omega^{-1} \cdot {}^\circ\mathrm{C}^{-1}(65^\circ\mathrm{C} - 20^\circ\mathrm{C})]$$
$$= 5.91\ \Omega$$

A Pt100 RTD (means a platinum RTD with $R_0 = 100\ \Omega$ at 0°C) has a resistance–temperature relationship described by the *Callendar–Van Dusen* equation:

$$R_T = R_0[1 + A'T + B'T^2 + C'(T - 100)T^3]_{(-200°C < T < 850°C)} \tag{2.8}$$

where A', B', and C' are *Callendar–Van Dusen* coefficients. Their values for different RTD standards are listed in Table 2.3 [3].

TABLE 2.3
Callendar–Van Dusen Coefficients Corresponding to Standard RTDs

Standard	**A'**	**B'**	**C' ($C' = 0$ for $T > 0°C$)**
DIN 43760	3.9080×10^{-3}	-5.8019×10^{-7}	-4.2735×10^{-12}
American	3.9692×10^{-3}	-5.8495×10^{-7}	-4.2325×10^{-12}
ITS-90	3.9848×10^{-3}	-5.8700×10^{-7}	-4.0000×10^{-12}

Source: From Measuring temperature with RTDs—A tutorial, Application Note 046, National Instruments Corporation, Austin, Texas, USA, 1996. With permission.

EXAMPLE 2.5

A Pt100 sensor is used to measure the temperature of a chamber. What is its resistance under a –80°C temperature? If the chamber's temperature is increased to +80°C, what is the sensor's new resistance value? Assume the American standard Pt100.

SOLUTIONS

From the second row in Table 2.3, $A' = 3.9692 \times 10^{-3}$ and $B' = -5.8495 \times 10^{-7}$. Thus:

For $T = -80°C$, $C' = -4.2325 \times 10^{-12}$:

$$R_{-80} = R_0[1 + A'T + B'T^2 + C'(T - 100)T^3]$$

$$R_{-80} = 100[1 + (3.9692 \times 10^{-3})(-80) + (-5.8495 \times 10^{-7})(-80)^2 + (-4.2325 \times 10^{-12})(-80 - 100)(-80)^3] = 67.83\ \Omega$$

For $T = 80°C$, $C' = 0$:

$$R_{80} = R_0[1 + A'T + B'T^2] = 100[1 + (3.9692 \times 10^{-3})(80) + (-5.8495 \times 10^{-7})(80)^2] = 131.38\ \Omega$$

RTDs

Common RTD Sensor Materials and Their Characteristics

Metal	Temperature Range (°C)	A ($\Omega \cdot \Omega^{-1} \cdot °C^{-1}$)	Comments
Platinum (Pt)	−240 ~ +850	0.00385	Good precision, broad temperature range
Nickel (Ni)	−80 ~ +260	0.00672	Low cost, limited temperature range
Copper (Cu)	−200 ~ +260	0.00427	Low cost, applied in measuring the temperature of electric motor and transformer windings
Molybdenum (Mo)	−200 ~ +200	0.00300 or 0.00385	Lower cost, alternative to platinum in the lower temperature ranges, ideal material for film-type RTDs

RTD Measurement

An RTD is a passive device, requiring a current to pass through to produce a measurable voltage. If the excitation current passing through an RTD is I_{ex}, and the output voltage across the RTD is V_{out}, then the measured temperature T (in °C) can be obtained by [11]

$$T = \frac{2(V_{out} - I_{ex}R_0)}{I_{ex}R_0[A + \sqrt{A^2 + 4B(V_0 - I_{ex}R_0)/(I_{ex}R_0)}]} \tag{2.23}$$

where R_0 is the resistance of the RTD at 0°C; A and B ($C = 0$ in this case when $T > 0$°C) are the *Callendar–Van Dusen Coefficients.*

EXAMPLE 2.9

A 0.15 mA excitation current is passed through a Pt100 RTD manufactured to a DIN 43760 standard. If the voltage output is 33 mV, what is the temperature measured (assume $T > 0°C$)?

Solution

From Table 2.3, $A = 3.9080 \times 10^{-3}$ and $B = -5.8019 \times 10^{-7}$, and $C = 0$, Thus

$$T = \frac{2(V_{out} - I_{ex}R_0)}{I_{ex}R_0[A + \sqrt{A^2 + 4B(V_0 - I_{ex}R_0)/(I_{ex}R_0)}]}$$

$$= \frac{2(0.033 - 0.15 \times 10^{-3} \times 100)}{(0.15 \times 10^{-3} \times 100)\left[3.9080 \times 10^{-3} + \sqrt{\begin{array}{l}(3.9080 \times 10^{-3})^2 + 4(5.8019 \times 10^{-7})(0.033 \\ -\ 0.15 \times 10^{-3} \times 100)/(0.15 \times 10^{-3} \times 100)\end{array}}\right]}$$

$$= 322.85°C$$

An RTD is a passive device, requiring a current to pass through to produce a measurable voltage. The excitation current also causes the RTD to heat internally (self-heating), which can result in a measurement error. Self-heating is typically specified as the amount of power that will raise the RTD's temperature by 1°C (in mW / °C). To minimize self-heating-caused error, the smallest possible excitation current (1 mA or less) should be used in measurement. The amount of self-heating also depends greatly on the medium in which the RTD is immersed. For example, an RTD can self-heat up to 100 times higher in still air than in moving water.

Platinum Temperature Sensor **Pt100**

Temperature range from -50 °C up to 0 °C:

$$R_t = R_o \cdot (1 + A \cdot t + B \cdot t^2 + C \cdot (t - 100\ °C) \cdot t^3)$$

Temperature range from 0 °C up to +600 °C:

$$R_t = R_o \cdot (1 + A \cdot t + B \cdot t^2)$$

$$A = 3.9083 \cdot 10^{-3}\ °C^{-1}$$
$$B = -5.775 \cdot 10^{-7}\ °C^{-2}$$
$$C = -4.183 \cdot 10^{-12}\ °C^{-4}$$

Resistance at 0 °C (R_0)	100 Ω
Temperature coefficient (0 °C up to +100 °C)	$3.85 \cdot 10^{-3}\ K^{-1}$
Tolerance classes according to DIN EN 60751	• F 0,15 (-30 °C - +300 °C) • F 0,3 (-50 °C - +500 °C)
Operating temperature range depending on lead material:	
AgPd5	-50 °C up to +400 °C
Pt	-50 °C up to +600 °C
Measurement current (DC) at 25 °C	1.0 mA
Maximal permissible peak current (DC) at 25 °C	3.0 mA
Insulation resistance	> 10 MΩ
Self-heating at 0 °C	< 0.2 K/mW
Thermal response time	
Flowing water (v = 0.2 m/s)	$T_{0.5} \leq 1.3$ s, $T_{0.9} \leq 5.0$ s
Flowing air (v = 1 m/s)	$T_{0.5} \leq 15$ s, $T_{0.9} \leq 50$ s

RTDs are constructed in two forms: *wire-wound* (Figure 2.11a) and *thin film* (Figure 2.11b). Wire-wound RTDs are made by winding a very fine strand of metal wire (platinum, typically 0.0005–0.0015 in. diameter). Thin-film RTDs are produced using *thin-film lithography* that deposits a thin film of metal (e.g., 1 μm platinum) onto a ceramic substrate through the *cathodic atomization* or *sputtering* process.

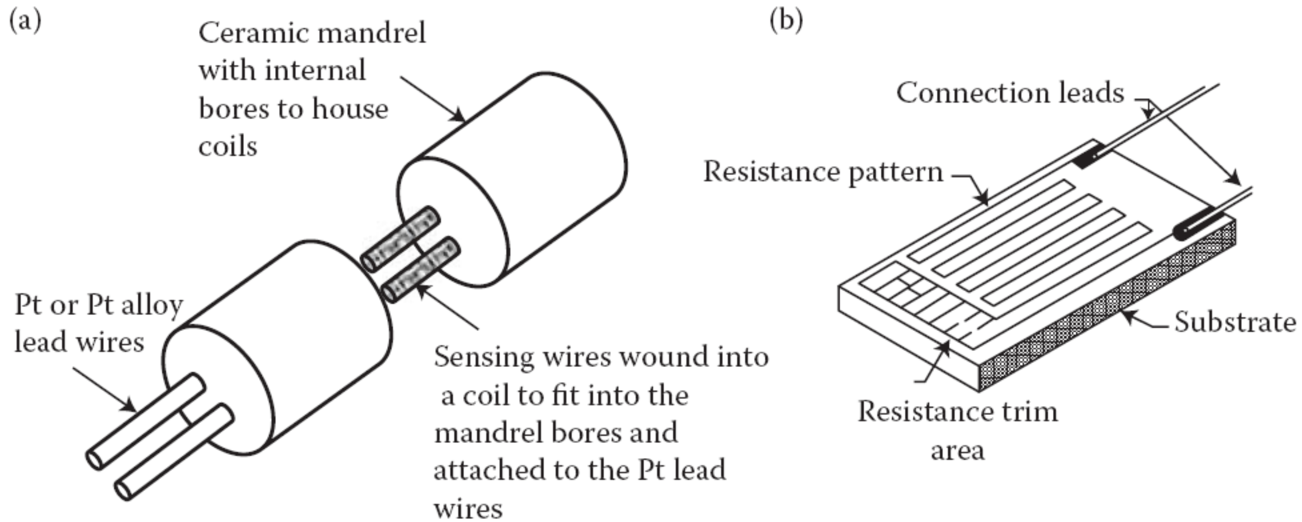


FIGURE 2.11 (a) Wire-wound RTD; (b) thin-film RTD. (Courtesy of *RdF Corporation*, Hudson, New Hampshire, USA.)

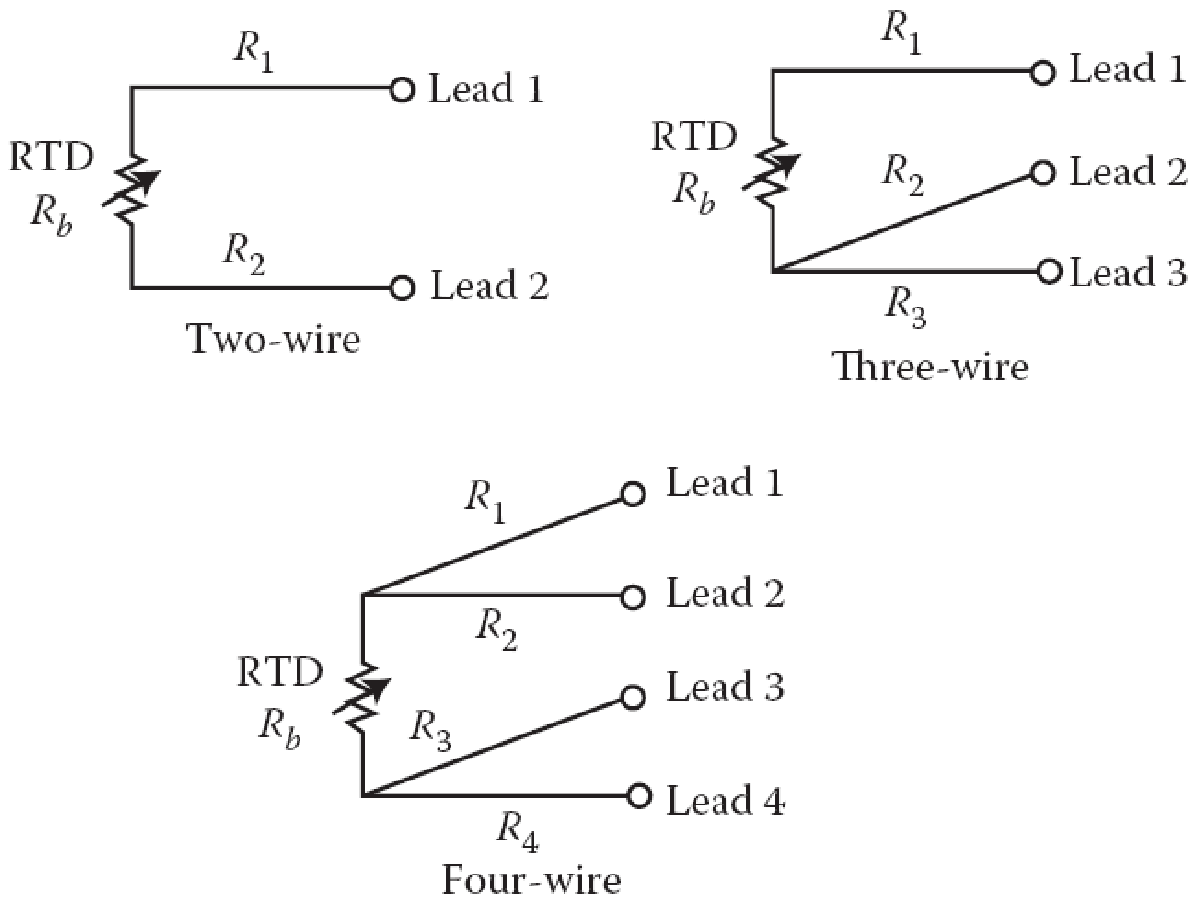


FIGURE 2.12 Two-, three-, and four-wire configurations.

Thermoresistive Effect for Semiconductors

In semiconductor materials, the valence electrons are bonded in covalent bonds with their neighbors. As temperature increases, thermal vibration of the atoms breaks up some of these bonds and releases electrons. These "free" electrons are able to move through the material under applied electric fields and the material appears to have a smaller resistance. Thus, electrical resistance R of semiconductor materials decreases as temperature T increases. The relationship between R and T is exponential (*Beta Equation*) [5]:

$$R_T = R_0 e^{\left[\beta\left(\frac{1}{T}-\frac{1}{T_0}\right)\right]} \tag{2.19}$$

where R_0 is the resistance at the reference temperature T_0 (in kelvin, K), usually 298 K (25°C), and β is the temperature coefficient (in K) of the material. Since resistance decreases as the temperature increases, β is a *negative temperature coefficient* (*NTC*). Most thermistors (a contraction of the words *therm*al and resi*stor*), made of semiconductor materials, are NTC sensors. A common resistance–temperature (R–T) curve for a 10 kΩ thermistor is shown in Figure 2.10 [6].

EXAMPLE 2.7

Find β of a thermistor whose values are $R_0 = 10\ \text{k}\Omega$ at $T_0 = 25°\text{C}$; $R_{50} = 3.3\ \text{k}\Omega$ at $T = 50°\text{C}$.

SOLUTION

From Equation 2.19

$$\beta = \frac{\ln R_T - \ln R_0}{1/T - 1/T_0} = \frac{\ln R_{50} - \ln R_{25}}{1/(273.15 + 50) - 1/(273.15 + 25)}$$

$$= \frac{\ln(3300) - \ln(10{,}000)}{1/(273.15 + 50) - 1/(273.15 + 25)} = 4272.66\ (\text{K})$$

The R–T curve of a thermistor is highly dependent upon its manufacturing process. Therefore, thermistor curves have not been standardized to the extent that RTD or thermocouple curves have been done. The thermistor curve, however, can be approximated with the *Steinhart–Hart equation*, which is an inverse and finer version of the *Beta Equation* 2.19 [7]:

$$T = \frac{1}{a + b\ln(R) + c\,[\ln(R)]^3} \tag{2.20}$$

where a, b, and c are constants, normally provided by manufacturers as part of the specification for each thermistor type, or alternatively provided as the R–T tables or curves. a, b, and c can also be determined by calibrating at three different temperatures and solving three simultaneous equations based on Equation 2.20.

The Steinhart–Hart Equation 2.20 has a third-order polynomial term, which provides excellent curve fitting for temperature spans within the range of –80°C to 260°C, and it has replaced the beta equation as the most useful tool for interpolating the NTC thermistor's R–T characteristics. If the full temperature range extends beyond the –80°C to 260°C range, the Steinhart–Hart equation can be used to fit a series of narrow temperature ranges, and then splice them together to cover the full range. Table 2.4 shows a typical thermistor data sheet (from *YSI Inc*., Yellow Springs, Ohio, USA, for its 44004 thermistor).

TABLE 2.4

Specifications of the 44004 Thermistor by *YSI Inc*.

Parameter	Specification
Resistance at 25°C	2252 Ω (100 Ω to 1 MΩ available)
Measurement range	–80°C to +120°C typical (250°C maximum)
Interchangeability (tolerance)	±0.1°C or ±0.2°C
Stability over 12 months	<0.02°C at 25°C, <0.25°C at 100°C
Time constant	<1.0 s in oil, <60 s in still air
Self-heating	0.13°C · $(mW)^{-1}$ in oil, 1.0°C · $(mW)^{-1}$ in air
Coefficients	$a = 1.4733 \times 10^{-3}$, $b = 2.372 \times 10^{-3}$, $c = 1.074 \times 10^{-7}$
Dimensions	Ellipsoid bead 2.5 mm × 4 mm

EXAMPLE 2.8

Three temperature points are selected for calibrating a thermistor. The resistances at each point are 7355 Ω at 0°C, 1200 Ω at 40°C, and 394.5 Ω at 70°C, respectively. (1) Find the constants *a*, *b*, and *c*. (2) What is the temperature if $R = 2152\ \Omega$?

SOLUTIONS

1. Plug the three sets of *R*s and *T*s into Equation 2.20:

$$(0 + 273) = \frac{1}{a + b \ln(7355) + c\,[\ln(7355)]^3}$$

$$\Rightarrow 273a + 2430.56b + 192660.03c - 1 = 0 \qquad (1)$$

$$(40 + 273) = \frac{1}{a + b\ln(1200) + c\,[\ln(1200)]^3}$$

$$\Rightarrow 313a + 2219.19b + 111557.09c - 1 = 0 \qquad (2)$$

$$(70 + 273) = \frac{1}{a + b\ln(394.5) + c\,[\ln(394.5)]^3}$$

$$\Rightarrow 343a + 2050.32b + 73262.01c - 1 = 0 \qquad (3)$$

Solving the three simultaneous Equations (1) through (3) yields

$$a = 1.47408 \times 10^{-3} \quad b = 2.3704159 \times 10^{-4} \quad c = 1.0839894 \times 10^{-7}$$

2. When $R = 2152\ \Omega$, the temperature is

$$T = \frac{1}{1.47408 \times 10^{-3} + 2.3704159 \times 10^{-4}\ln(2152) + 1.0839894 \times 10^{-7}\ [\ln(2152)]^3}$$

$$= 299.21\ \mathrm{K}\left(\text{or } 26.21°\mathrm{C}\right)$$

انواع ترمیستور

۱- ترمیستور با ضریب تغییرات حرارتی منفی (NTC)

۲- ترمیستور با ضریب تغییرات مثبت (PTC)

- Thermally sensitive silicon resistors (*silistors*)
- Switching PTC thermistors

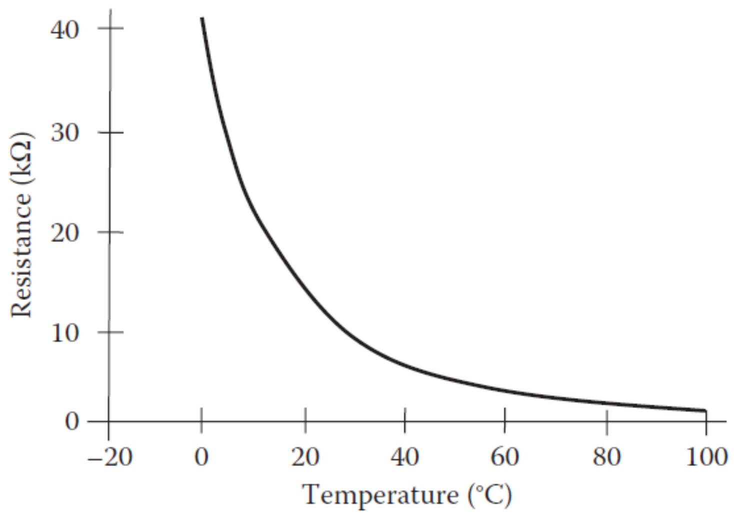


FIGURE 2.10 An NTC thermistor R–T curve.

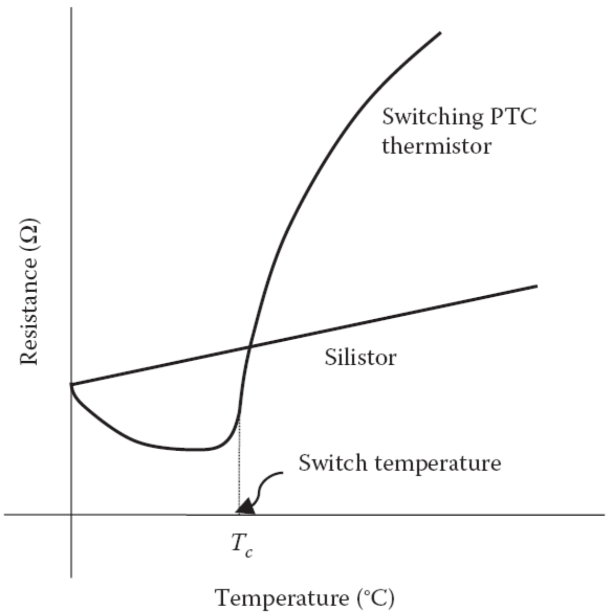


FIGURE 2.14 R–T curves of silistor and switching PTC thermistors.

❖ NTC thermistors are more commonly used than PTC type, especially in temperature measurement applications. NTC thermistors are made using basic ceramics technology and semiconductor metal oxide materials (e.g., oxides of manganese, nickel, cobalt, iron, copper, and titanium). In some thermistors, the decrease in resistance is as great as 6% for each 1°C of temperature increase although 1% changes are more typical. NTC thermistors can provide good accuracy and resolution when measuring temperatures between −100°C and +300°C. If inserted into a Wheatstone bridge, a thermistor can detect temperature changes as small as ±0.005°C.

Part No.	R25℃ (Ω)	Max.Steady State Current (A)	Approx.R @ Max.Cur (Ω)	Dissipation factor (δ)(mW/℃)	Thermal time constant (s)
NTC-5D5	5	1	0.353	6	20
NTC-10D5	10	0.7	0.771	6	20
NTC-60D5	60	0.5	1.878	6	18
NTC-200D5	200	0.1	6.259	6	18

- Silistors exhibit a fairly uniform PTC (about +0.0077°C−1) through most of their operational range, but can also exhibit an NTC region at temperatures higher than 150°C. This type of thermistors is often used for temperature compensation of silicon semiconducting devices in the range of −60°C to +150°C.

- The switching PTC thermistors are made from polycrystalline ceramic materials that are normally highly resistive but become semiconductive by adding dopants. They are often manufactured using compositions of barium, lead, and strontium with additives such as yttrium, manganese, tantalum, and silica. The R–T curves of switching PTC thermistors exhibit very small NTC regions until they reach a critical temperature *Tc*—"*Curie*," "*Switch*" or "*Transition*" *temperature*. After *Tc*, the curve exhibits a rapidly increasing PTC resistance. These resistance changes can be as much as several orders of magnitude within a temperature span of a few degrees.


Thanks for Attention